See from the last page

The mean recurrence time,

$$\mu_{00} = \frac{1}{P_0} = \frac{65}{32}$$

$$\mu_{11} = \frac{1}{P_1} = \frac{65}{33}$$

~~fios~~ $P^4 = \begin{bmatrix} 0.5065 & 0.4934 \\ 0.4785 & 0.5214 \end{bmatrix}$

Now first passage time,

$f_{01}^{(4)} = P_{01}^{(4)} - f_{01}^{(1)} P_{11}^{(3)} - f_{01}$

$f_{01}^{(1)} = P_{01} = \frac{3}{10}$

$f_{01}^{(2)} = P_{01}^{(2)} - f_{01}^{(1)} P_{11} = \frac{93}{220} - \frac{3}{10} \times \frac{39}{55} = \frac{21}{100}$

$f_{01}^{(3)} = P_{01}^{(3)} - f_{01}^{(1)} P_{11}^{(2)} - f_{01}^{(2)} P_{11}$

$= \frac{2289}{4840} - \frac{3}{10} \times \frac{357}{605} - \frac{21}{100} \times \frac{39}{55}$

$= \frac{147}{1000}$

$f_{01}^{(4)} = P_{01}^{(4)} - f_{01}^{(1)} P_{11}^{(3)} - f_{01}^{(2)} P_{11}^{(2)} - f_{01}^{(3)} P_{11}$

$= 0.4934 - \frac{3}{10} \times \frac{3603}{6655} - \frac{21}{100} \times \frac{357}{605} - \frac{147}{1000} \times \frac{39}{55}$

$= 0.1028$

$\therefore$ The required first passage time $f_{01}^{(4)} = 0.1028$.

Roll: 361

From transition probability matrix P, the equations,

$$P_0 = \frac{7}{10} P_0 + \frac{16}{55} P_1$$

$$1 = P_0 + P_1$$

Solving these equations we get,

$$P_0 = \frac{32}{65} \quad \text{and} \quad P_1 = \frac{33}{65}$$

which are the required steady state probabilities.

Now,

~~$P^2 = \begin{bmatrix} 0.5772 & 0.4227 \end{bmatrix}$~~

$$P^2 = \begin{bmatrix} \frac{127}{220} & \frac{93}{220} \\ \frac{248}{605} & \frac{357}{605} \end{bmatrix}$$

$$P^3 = \begin{bmatrix} \frac{2551}{4840} & \frac{2289}{4840} \\ \frac{3052}{6655} & \frac{3603}{6655} \end{bmatrix}$$

Answer to Q no. 5

Given transition matrix,

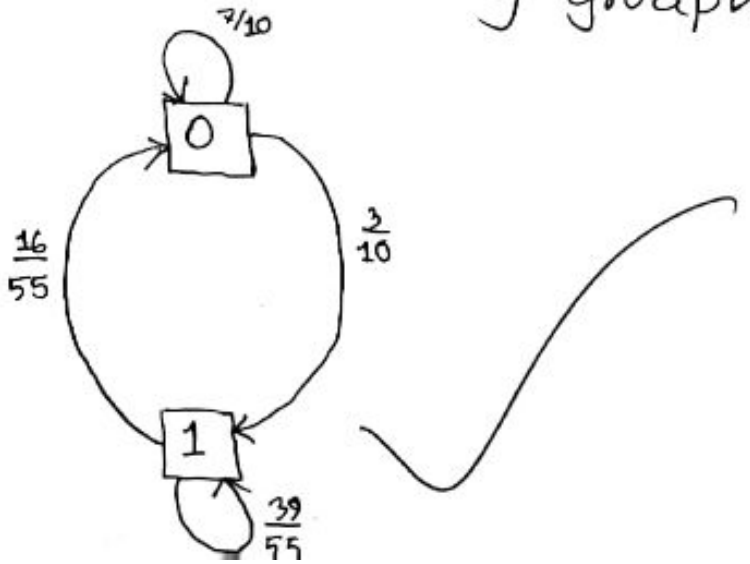
$$TM = \begin{array}{c} \\ 0 \\ 1 \end{array} \begin{array}{c} \begin{array}{cc} 0 & 1 \end{array} \\ \begin{bmatrix} 224 & 96 \\ 96 & 234 \end{bmatrix} \end{array}$$

∴ The transition probability matrix,

$$P = \begin{array}{c} \\ 0 \\ 1 \end{array} \begin{array}{c} \begin{array}{cc} 0 & 1 \end{array} \\ \begin{bmatrix} \frac{224}{320} & \frac{96}{320} \\ \frac{96}{330} & \frac{234}{330} \end{bmatrix} \end{array}$$; [∵ Sum along row 0 = 224+96 = 32
Sum along row 1 = 234+96 = 330

$$= \begin{array}{c} \\ 0 \\ 1 \end{array} \begin{array}{c} \begin{array}{cc} 0 & 1 \end{array} \\ \begin{bmatrix} 0.7 & 0.3 \\ \text{[illegible]} & \text{[illegible]} \end{bmatrix} \end{array} \quad \begin{array}{c} \\ 0 \\ 1 \end{array} \begin{array}{c} \begin{array}{cc} 0 & 1 \end{array} \\ \begin{bmatrix} \frac{7}{10} & \frac{3}{10} \\ \frac{16}{55} & \frac{39}{55} \end{bmatrix} \end{array}$$

The transition probability graph is as follows:



Mean time of failure,

$$\mu = \gamma + \delta \Gamma\left(1 + \frac{1}{\beta}\right) = 0 + 100 \cdot \Gamma\left(1 + \frac{1}{1/2}\right)$$

$$= 100 \cdot \Gamma 3$$

$$= 200$$

∴ The mean time of failure is 200 hours.

Variance ~~of time of~~ of time of failure,

$$\sigma^2 = \delta^2 \left[\Gamma\left(1 + \frac{2}{\beta}\right) - \left\{\Gamma\left(1 + \frac{1}{\beta}\right)\right\}^2\right]$$

$$= 100^2 \left[\Gamma\left(1 + \frac{2}{1/2}\right) - \left\{\Gamma\left(1 + \frac{1}{1/2}\right)\right\}^2\right]$$

$$= 100^2 \left[\Gamma 5 - (\Gamma 3)^2\right] = 100^2 (24 - 4) = 100^2 \times 20$$

$= 2000$ ×

∴ Standard deviation of time of failure,

$$\sigma = \sqrt{2000} = 44.7214$$

$$\sigma = 447$$

Answers to Q no. 4

Let,

X = time to failure density for a small computers system

Here, X follows Weibull distribution with parameters $\gamma = 0$, $\beta = \frac{1}{2}$ and $\delta = 100$.

∴ The probability density function,

$$f(x) = \begin{cases} \frac{1}{200}\left(\frac{x}{100}\right)^{\frac{1}{2}-1} e^{-\left(\frac{x}{100}\right)^{\frac{1}{2}}} & ; x \geq 0 \\ 0 & ; \text{otherwise} \end{cases}$$

and cumulative density function,

$$F(x) = 1 - e^{-\left(\frac{x}{100}\right)^{\frac{1}{2}}} \quad ; x \geq 0$$

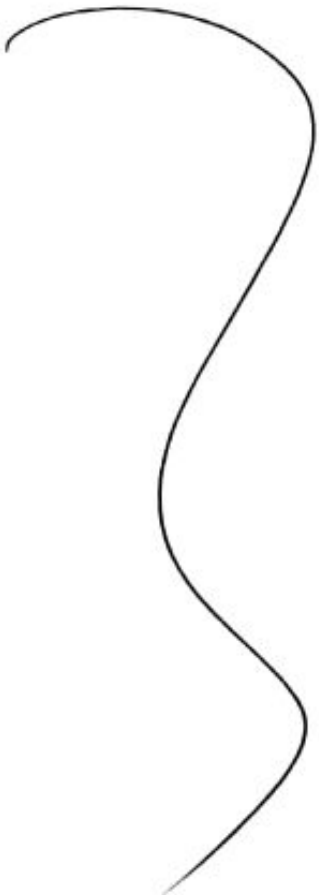
Now, the fraction of these units will survive to 500 hours is,

$$1 - F(500) = e^{-\left(\frac{500}{100}\right)^{\frac{1}{2}}} = 0.1069$$

If $\alpha = 1$ and $\beta = \frac{1}{2}$ then Gamma distribution becomes exponential distribution.

Some uses of Gamma distribution are as follows:

1) To calculate load of servers.



Here, $n = 20$

$\therefore$ Rank correlation, $r_s = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n(n^2-1)}$

$= 1 - \frac{6 \times 2160}{20(20^2-1)}$ → 2340

$= -0.6241$

$= -0.76$

Here, the value of r_s is negative.

$\therefore$ This particular EEG output is inversely correlated with age.

Answers to Q no. 3

If X is a continuous random variable and α and β are two parameters then X follows Gamma distribution with pdf,

$$f(x, \alpha, \beta) = \begin{cases} \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta} & ; x \geq 0 \\ 0 & ; \text{otherwise} \end{cases}$$

Answers to Q no. 2

ID	Age	Rank (Age)	Egg	Rank (Egg)	d	d^2
1	20	1	96	18	-17	289
2	21	2	75	15	-13	169
3	22	3	95	17	-14	196
4	24	4	100	20	-16	256
5	27	5	99	19	-14	196
6	30	6	65	7	-1	1
7	31	7	64	6	1	1
8	33	8	70	12	-4	16
9	35	9	65	16	-7	49
10	38	10	74	14	-4	16
11	40	11	68	10	1	1
12	42	12	66	8	4	16
13	44	13	71	13	0	0
14	46	14	62	4	10	100
15	48	15	69	11	4	16
16	51	16	54	2	4	16
17	53	17	63	5	12	144
18	55	18	52	1	17	289
19	58	19	67	9	10	100
20	60	20	55	3	17	289
						$\Sigma d^2 = 2160$

$\Sigma d^2 = 2340$

is not needed for ~~del~~ calculation.

ii) Rank correlation can be either positive or negative. But point biserial correlation is positive (~~abs~~ absolute value is taken if found negative).

The situations where the correlations are used are -

i) To find the relation of the marks obtained by students in two different courses rank correlation is used.

ii) To find the relation of the marks obtained by male and female students in a course point biserial correlation is used.

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OBTAINED MARK 25

TUTORIAL ANSWER SCRIPT

Student's ID No: 361 **Name:** Mahiyat Tanzim

Course Code: CSE-251 **Course Title:** Probability and Statistics

Tutorial Examination No: 03 **Date:** 09/02/23 **Signature of Course Teacher:** _____

Answer to Q no. 1

Rank correlation is the linear relationship among two different types of data on the basis of rank. $r_s = 1 - \frac{6 \Sigma d^2}{n(n^2-1)}$

The differences between rank correlation and point biserial correlation are as follows:

1) In rank correlation, both types of data are non-binomial (Bernoulli) ~~But in~~ and relation is determined on the basis of rank. But in point biserial correlation, one is binomial (0,1) and the others is non-binomial and rank